



LECTURE 01 : INTRODUCTION TO C++



What is C++?

C++ is a programming language used to write computer programs.



Simple Definition

- A language that humans can write
- Computer understands it and follows instructions
- Used to create software, games, apps



Created By

- Bjarne Stroustrup in 1979
- Built on top of C language
- Made to be fast and powerful



Why Do We Need C++?



1. Speed and Performance

C++ programs run very fast.



Used in:

- Games (GTA, Call of Duty, Fortnite)
- Operating Systems (Windows, macOS)
- Browsers (Chrome, Firefox)



2. Control Over Computer

C++ gives direct control over:

- Memory
- Hardware
- CPU



3. Real-World Uses

Field	Examples
Games	Unreal Engine, Unity
Systems	Windows, Linux
Apps	Photoshop, Microsoft Office
Databases	MySQL, MongoDB

Field	Examples
Browsers	Chrome (V8 engine)
Embedded	Car systems, Drones
Finance	Trading systems

4. Foundation Language

Learning C++ helps you understand:

- How computers work
- Memory management
- Other languages become easier

Binary Number System

? What is Binary?

Binary = Number system with only 2 digits: 0 and 1

- Computers only understand 0 and 1
- 0 = OFF (no electricity)
- 1 = ON (electricity flowing)

⚡ Why Binary?

Computers use electricity:

- 0 = Low voltage (OFF)
- 1 = High voltage (ON)

Easy for computers to understand ON/OFF states.

Decimal vs Binary

Decimal	Binary	Why?
0	0	Zero
1	1	One
2	10	Two
3	11	Three
4	100	Four
5	101	Five

Decimal	Binary	Why?
10	1010	Ten

Decimal to Binary Conversion

 **Method:** Divide by 2

Steps:

1. Divide number by 2
2. Write remainder
3. Divide quotient by 2
4. Repeat until quotient = 0
5. Read remainders from **bottom to top**

Example 1: Convert 13 to Binary

Step 1: $13 \div 2 = 6$, remainder = 1
Step 2: $6 \div 2 = 3$, remainder = 0
Step 3: $3 \div 2 = 1$, remainder = 1
Step 4: $1 \div 2 = 0$, remainder = 1

Read from bottom to top: 1101

Answer: 13 = 1101

Example 2: Convert 25 to Binary

$25 \div 2 = 12$, remainder = 1
 $12 \div 2 = 6$, remainder = 0
 $6 \div 2 = 3$, remainder = 0
 $3 \div 2 = 1$, remainder = 1
 $1 \div 2 = 0$, remainder = 1

Read from bottom to top: 11001

Answer: 25 = 11001

Example 3: Convert 8 to Binary

$8 \div 2 = 4$, remainder = 0
 $4 \div 2 = 2$, remainder = 0

2 ÷ 2 = 1, remainder = 0
1 ÷ 2 = 0, remainder = 1

Read from bottom to top: 1000

Answer: 8 = 1000

 Quick Practice

Decimal	Binary
5	101
7	111
10	1010
15	1111
20	10100

 Binary to Decimal Conversion

 Method: Powers of 2

 Steps:

- 1. Write powers of 2 from right to left (start with 2⁰)
- 2. Multiply each binary digit with its power
- 3. Add all results

 Powers of 2 Table

Position	Power	Value
0 (rightmost)	2 ⁰	1
1	2 ¹	2
2	2 ²	4
3	2 ³	8
4	2 ⁴	16
5	2 ⁵	32

Position	Power	Value
6	2^6	64
7	2^7	128

 Example 1: Convert 1101 to Decimal

Binary: 1101

Power: 2³ 2² 2¹ 2⁰

Value: 8 4 2 1

Step 1: 1 × 8 = 8

Step 2: 1 × 4 = 4

Step 3: 0 × 2 = 0

Step 4: 1 × 1 = 1

Total: 8 + 4 + 0 + 1 = 13

Answer: 1101 = 13

 Example 2: Convert 1010 to Decimal

Binary: 1010

Power: 2³ 2² 2¹ 2⁰

Value: 8 4 2 1

Step 1: 1 × 8 = 8

Step 2: 0 × 4 = 0

Step 3: 1 × 2 = 2

Step 4: 0 × 1 = 0

Total: 8 + 0 + 2 + 0 = 10

Answer: 1010 = 10

 Example 3: Convert 11001 to Decimal

Binary: 11001

Power: 2⁴ 2³ 2² 2¹ 2⁰

Value: 16 8 4 2 1

Step 1: 1 × 16 = 16

Step 2: 1 × 8 = 8

Step 3: 0 × 4 = 0

Step 4: $0 \times 2 = 0$

Step 5: $1 \times 1 = 1$

Total: $16 + 8 + 0 + 0 + 1 = 25$

Answer: $11001 = 25$

✓ Example 4: Convert 1111 to Decimal

Binary: 1 1 1 1
Power: 2^3 2^2 2^1 2^0
Value: 8 4 2 1

Step 1: $1 \times 8 = 8$

Step 2: $1 \times 4 = 4$

Step 3: $1 \times 2 = 2$

Step 4: $1 \times 1 = 1$

Total: $8 + 4 + 2 + 1 = 15$

Answer: $1111 = 15$

Quick Practice

Binary	Decimal
101	5
111	7
1000	8
10100	20
11111	31

Summary

★ What is C++?

- Programming language
- Fast and powerful
- Used in games, systems, apps

⚡ Why Binary?

- Computers understand only 0 and 1
- 0 = OFF, 1 = ON

- Easy for electronics

Decimal to Binary

- Divide by 2 repeatedly
- Write remainders
- Read bottom to top

Binary to Decimal

- Use powers of 2
- Multiply and add
- Get decimal number

Quick Reference

Decimal to Binary Steps

1. Divide by 2
2. Write remainder
3. Repeat
4. Read reverse

Binary to Decimal Steps

1. Write powers of 2
2. Multiply each digit
3. Add all results

Common Conversions

Decimal → Binary

0 → 0
1 → 1
2 → 10
3 → 11
4 → 100
5 → 101
10 → 1010
15 → 1111

Practice these conversions daily! 

Where C++ is used

- 1: Database
- 2: NodeJS

- 3: Javascript
- 4: Game Development
- 5: HFT
- 6: Machine Learning Libraries(Tensorflow)

C++ Code Execution

```
#include<iostream>    // Header file
using namespace std;  // std namespace use karne ke liye

int main() {

    int a = 10;        // Variable declaration

    cout << a;         // Output

    return 0;          // Program successfully run hua
}
```

Important Notes

- cin and cout C++ ka part hai
- `#include<iostream>` C++ ki library hai
- iostream ka meaning input output stream hota hai
- C++ me 1000+ libraries hoti hai
- Jis library ka use karna ho usse include kar lete hai
- `using namespace std;`
likhne se cout, cin ke sath std automatically include ho jata hai
- `std::cout << a;`
ka matlab cout std ke andar hai

Code Execution Flow

C++ Code → Compiler → Machine Code → Computer

Compiler C++ code ko machine code me convert karta hai
jisse computer samajh pata hai.

1. Variables

What is a Variable?

A container that stores data in memory.

```
int age = 25;
```

- `int` = type of container
- `age` = name of container
- `25` = value stored

Rules for Variable Names

Allowed

```
int age;  
int student_name;  
int rollNo123;  
int _value;
```

Not Allowed

```
int 123roll;      // Can't start with number  
int student-name; // No hyphens  
int int;          // Can't use keywords  
int my name;      // No spaces
```

Declaring Variables

```
// Method 1: Declare then assign  
int age;  
age = 25;  
  
// Method 2: Declare and assign together  
int age = 25;  
  
// Method 3: Multiple variables  
int a = 5, b = 10, c = 15;
```

2. Data Types



Basic Data Types

Data Type	Used For	Size	Example
int	Whole numbers	4 bytes	int age = 25;
double	Decimal numbers	8 bytes	double price = 99.99;
char	Single character	1 byte	char grade = 'A';
bool	True/False	1 byte	bool isPassed = true;
string	Text	Varies	string name = "Rohit";



Examples

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    int age = 25;
    double height = 5.9;
    char grade = 'A';
    bool isStudent = true;
    string name = "Rohit";

    cout << "Name: " << name << endl;
    cout << "Age: " << age << endl;
    cout << "Height: " << height << endl;
    cout << "Grade: " << grade << endl;
    cout << "Student: " << isStudent << endl;

    return 0;
}
```



Important Notes

For char : Use single quotes 'A'

```
char letter = 'A'; //  Correct
char letter = "A"; //  Wrong
```

For `string`: Use double quotes `"text"`

```
string name = "Rohit"; //  Correct
string name = 'Rohit'; //  Wrong
```



3. Strings



Basic String Operations

```
#include <iostream>
#include <string>
using namespace std;

int main() {

    string name = "Rohit";

    // Length of string
    cout << name.length() << endl; // 5

    // Combine strings
    string firstName = "Rohit";
    string lastName = "Kumar";
    string fullName = firstName + " " + lastName;

    cout << fullName << endl; // Rohit Kumar

    // Access characters (starts from 0)
    cout << name[0] << endl; // R
    cout << name[1] << endl; // o

    return 0;
}
```



String Input with Spaces



Problem

```
string name;

cin >> name; // Only reads one word!

// Input: "Rohit Kumar"
// Stores: "Rohit"
```

Solution

```
string name;  
  
getline(cin, name); // Reads entire line  
  
// Input: "Rohit Kumar"  
// Stores: "Rohit Kumar"
```